Choose a motion method

antocciphysics

UNIT 1 / DAY 8 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose a motion relationship and justify its assumptions.

SUCCESS CRITERIA - I can...

1. Choose a relationship using the known quantities.
2. Show linked calculations with units and directions.
3. State when constant velocity or acceleration applies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

x_i=-3±0.5 m, v=5±0.1 m/s and exact t=7 s. Give central final position, its conservative interval and full width. Name an assumption.

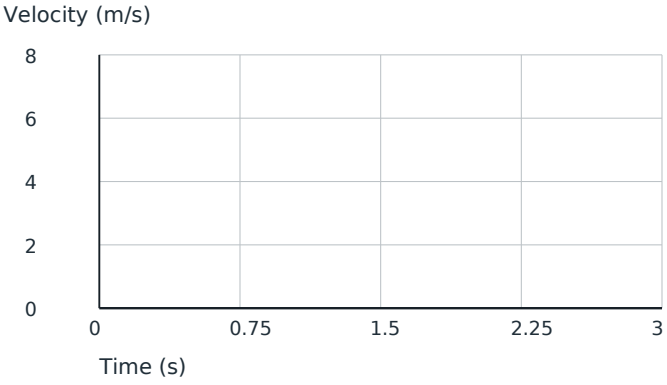
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_f = v_i + a\,\Delta t$; $\Delta x = v_i\,\Delta t + \frac{1}{2}a(\Delta t)^2$
$v_f^2 = v_i^2 + 2\,a\,\Delta x$.
Constant acceleration; i = initial, f = final. Choose the equation containing your knowns and unknown.

Worked reasoning: From rest, a cart accelerates uniformly at 2 m/s² for 3 s. Final velocity=6 m/s; average velocity=(0+6)/2=3 m/s; displacement=3×3=9 m.

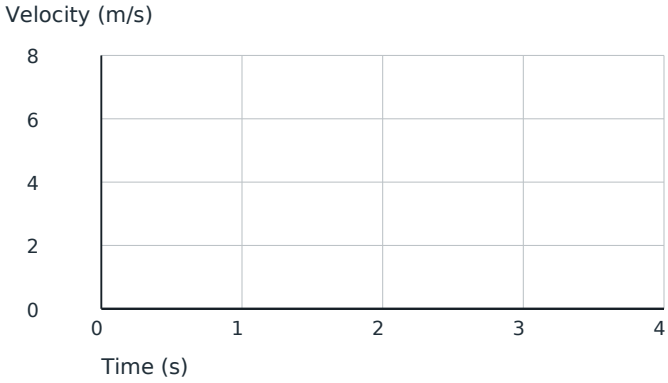
Your turn: Complete the cart's velocity-time graph. Shade the triangular area under the line and use it to check the 9 m displacement.



Marzano check - circle: ① ② ③

03 Solve and check

A cart starts at the -3 m mark on a test track, moving at +2 m/s. It accelerates at a constant +1 m/s² for 4 s. Draw its velocity-time graph, then find final velocity, displacement and final position.



Marzano check - circle: ① ② ③

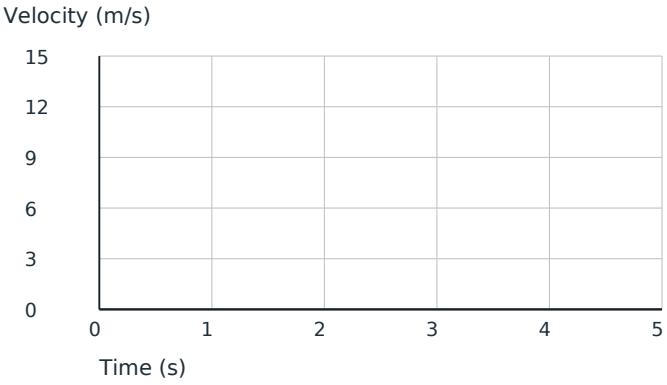
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Continue down each column.

04 No time given

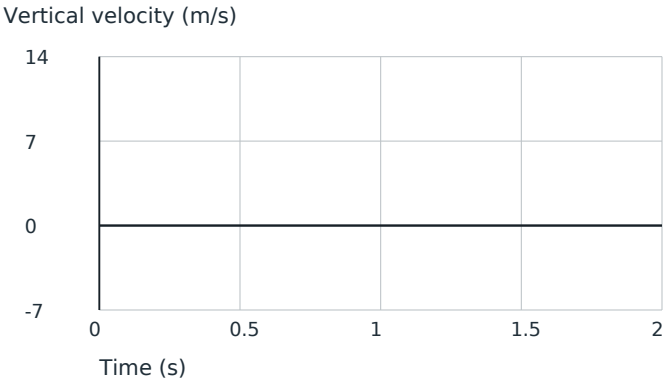
A cyclist brakes from 12 m/s to rest with constant acceleration -3 m/s². Forward is positive. Find stopping distance without first calculating time. Then find the stopping time and check with a velocity-time graph.



Marzano check - circle: 1 2 3

05 At the top

You toss a ball straight upward at 14 m/s. Ignore air resistance and use g = 9.8 m/s². Draw velocity and acceleration arrows at release and at the highest point. Find the maximum rise and time to the top. Is acceleration zero there?



Marzano check - circle: 1 2 3

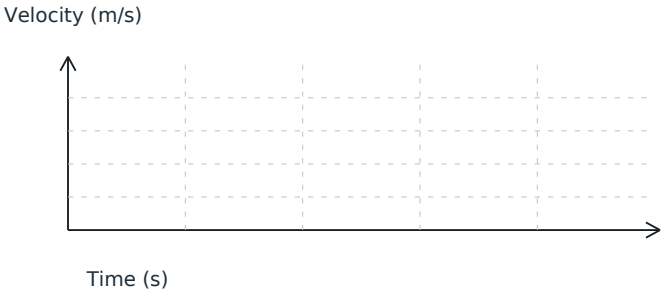
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Continue down each column.

06 Review: connect an earlier idea

Velocity changes from -3 to -15 m/s in 3 s, then stays at -15 m/s for 7 s. Find acceleration and describe the speed trend in each interval.
Sketch the velocity-time model; label the given values.



Marzano check - circle: 1 2 3

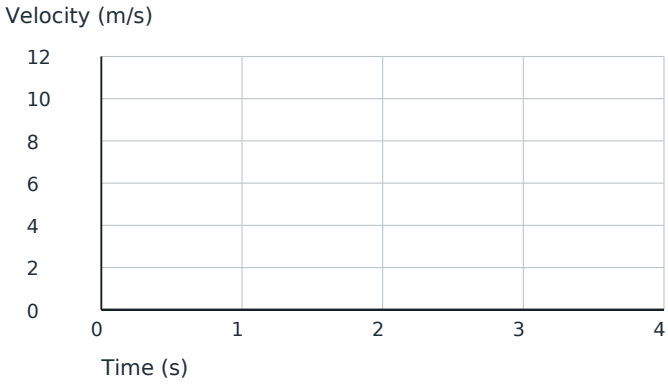
07 Exit check

A car starts at +12 m/s and brakes with a=-3 m/s² until it stops. A student extends the equation to 8 s and reports v=-12 m/s. Explain what assumption fails and give stopping time/distance.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A cart starts at 2 m/s and travels 24 m in 4 s with uniform acceleration. Solve for acceleration in two ways and explain why a constant-speed method fails.



Marzano check - circle: 1 2 3